

BRAIN TUMOR DETECTION USING CONVOLUTIONAL NEURAL NETWORKS ON MRI IMAGES

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Abstract—Diagnosis of brain tumor has been taken to be one of the greatest applications of medical imaging, as it assists in enhancing the survival rate as long as it is diagnosed at early stages of growth. The work of examining the Magnetic Resonance Imaging scan pictures is a time consuming activity since it would need examination by a radiologist owing to tumor variations. The earlier techniques applied in machine learning entail feature identification, which makes them weak.

With these difficulties in mind, an automatic system of brain tumor detection on the basis of CNN of the MRI images is offered in the present paper. The given system also relies on the transfer learning mechanism of using light MobileNetV2 model, due to which the system can successfully identify the predominant features within the pictures and classify the MRI pictures into the tumor and non-tumor categories. The experimental study proves the hypothesis that the suggested system can effectively detect brain tumors in less time of computation on the MRI images. Thus, it is undoubted that the suggested system can be successfully used as a CAD tool when it is combined with radiologists.

Index Terms—Brain tumor detection, MRI, Convolutional Neural Network, MobileNetV2, Transfer learning, Medical image analysis.

I. INTRODUCTION

Brain tumors are one of the most serious and threatening neurological complications that should be identified in time and correctly in order to guarantee higher survival rates among patients. The early intervention can lead to severe complications like the irreversible neurological damage and decreased treatment outcomes. Magnetic Resonance Imaging (MRI) is one of the most common forms of imaging used in the diagnosis of brain tumors due to its high soft-tissue contrast, high spatial resolution and the fact that it is non-invasive as it does not involve the use of ionizing radiation. However, interpretation of MRI scans manually is a time

consuming and experience, process that is likely to be subject to inter-observer variability.

The medical imaging information has been rising at a fast rate and there has been an increased demand to have computer-aided diagnosis (CAD) systems that are automated so that they can help the radiologists to make prompt and stable decisions. Brain tumor has been applied to the classical machine learning algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random forests and Logistic Regression. However, these techniques require a lot of reliance on manually produced techniques of feature extraction which cannot reflect the heterogeneous and complex nature of tumor tissues that can be seen on MRI images.

The recent years have seen the revolution of medical image analysis with the help of Convolutional Neural Networks (CNNs) and deep learning. The positive side of CNNs is the fact that they can learn hierarchical and discriminative features without feature engineering since they learn using raw images. This has led to an enormous step forward in precision in tasks such as tumor detection, classification and segmentation. Despite their high effectiveness, majority of deep CNN architectures require numerous computations and huge amount of memory and training time as well as are less viable to real time implementation or application in resource limited clinical operating systems.

In this paper, the researcher will address these concerns by addressing the use of MobileNetV2 that is a lightweight model of deep learning that can be effectively used to conduct computations with minimal parameters and the processing requirements are less. Learning can be transferred in the model, and it leverages the information that is learned in large-scale image databases and has the potential to converge quicker and offer superior results even with low quantities of medical training data.

The general objective of this project is to design an appropriate and accurate system of brain tumor classification based on MRI image in a CNN model. The aim of the proposed system is to achieve high classification and low cost of computation which shall prove to be relevant in the practical implementations in the clinical practice. The system has the ability to assist the radiologists detect tumors at the earliest possible level and it can assist in augmenting the diagnostic procedures since it will present an automated screening device.

The rest of the paper has the following structure: Section II, related work, brain tumor detection with deep learning, Section III, the proposed methodology, Section IV, the results of the experiment and performance evaluation, and Section V, paper conclusion and the future research directions.

II. LITERATURE REVIEW

Magboo et al. developed a multiclass model of brain tumor of MRI images based on the convolutional neural network (CNN) framework of the images on the basis of the images obtained by MRI b1. The model was not inferior in its precision and based on the small size of the dataset, the extent to which the model could be applied to the unknown information was also limited.

The authors made comparisons of CNN-based models with the traditional machine learning classifiers, which are the Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) classifiers to the CNN-based models b2. They discovered CNNs as being far superior but the model overfitted because of a lack of many training samples.

Ahmed applied deep convolutional networks to transfer learning in determining and segmenting brain tumor automatically [3]. Even though this strategy increased the accuracy of the detecting, it involved a lot of computational capacity and a multi sequence MRI image, therefore not achievable in a low resource setting.

Badza et al. have developed certain CNNs and compared them to machine learning algorithms in brain tumor classification [4]. This study was not done of the entire three dimensional tumor images but the two dimensional cross sections of the MRI images thus not so much in confirming the third dimensional aspect of the tumor [4].

Transfer learning principle has been applied by Singh on brain tumor analysis whereby it has applied deep learning models of VGG16 and ResNet b5. It was also the computationally expensive model that could not be run in real time and even on a CPU but at the same time it was also accurate.

Kumar et al. suggested CNN architecture of benign and malignant tumors of the brain classification [6]. The problem of the absence of the balance of classes is also current in this data set; it complicates the emergence of the real results about the measures of the performance of models.

Verma was able to develop a light CNN, which was used in determining brain tumours in real-time [7]. Even though the efficiency was improved, deterioration of performance was realized in finer tumor patterns.

The optimized CNN structure with the improved regularization methods was presented as an extension of the latest works of Ahmed et al. to raise the rates of accuracy of the model that is a system of the network and its parameters and configurations that contribute to the better rates of image recognition and the higher levels of the accuracy of the given system at its core (Ahmed et al. 2018, b8). The model made the training more complicated even in situations where it led to improved performance.

On the whole, the literature indicates that the techniques that rely on the idea of deep learning are precise, but the majority of the currently existing systems are highly costly both in terms of computational cost and deployability. The limitations encourage the suggested lightweight CNN model of MobileNetV2 to learn brain tumor detection capabilities successfully and efficiently.

III. EXISTING SYSTEM

A significant number of the current brain tumor detection systems implement deep learning to process MRI brain scans. Convolutional Neural Networks (CNNs) have become very popular due to the capability of detecting complex features in the form of medical images automatically. Various solutions adopt transfer learning with pre-trained models like VGGNet, ResNet, AlexNet, and DenseNet to enhance classification results and also decrease the training time and the requirement of very large medical datasets.

Other systems are binary-classification based in deciding whether a tumor exists, and the more developed ones do multi-classification in order to identify a particular type of tumor. Moreover, there are researches that combine tumor segmentation techniques with classification to enhance the accuracy by addressing the affected tissues. These methods have recorded great results and have assisted a great deal in computer-aided diagnosis of neuroimaging.

Most of the current models are however computationally intense and require powerful hardware to be used and is thus restricted by the resource constraint in medical environments or in real-time usage. Other techniques also have a problem with insufficient diversity in data, which results in a low level of generalization on unseen MRI scans. These issues show that there is a necessity to develop more efficient and lightweight deep learning models that would not compromise quality but reduce computing needs.

A. Existing Approach

In the specified approach MRI scans of the brain are being scanned and inputted into the system. The images are processed and then trained and as such, the images become clearer and devoid of inconsistencies. The CNN paradigm subsequently learns useful patterns of the MRI scans. This system will be founded on the pattern that has been learned on which it predicts the presence or absence of the tumor.

B. Limitations

However, in spite of the effectiveness of such systems, there are certain limitations to their practice. The models are very

large and thus use more computing power that is aided by the application of the GPU computing. The models will require more time to be calculated and this means that the patients will not be subject to real-time diagnosis because of the use of this technology. This is because the resources at the smaller hospitals and health facilities would not be able to sustain the application of this technology.

IV. PROPOSED SYSTEM

A. Overview

The proposed system is an automated and effective system of brain tumor detection using the MRI images with deep learning. It takes advantage of transfer learning using the lightweight MobileNetV2 to obtain correct classification at a lower level of computational complexity. Image resizing, image normalization are preprocessing steps to standardize the input data, and then the data is augmented to enhance the generalization of the model. The pretrained MobileNetV2 model is a feature extractor that picks out key spatial patterns of MRI scans. Above this another set of custom classification layers with a fully connected layer, dropout regularization, and a sigmoid output layer are added to classify tumors in a binary way. The model is trained and validated with the help of some structured data division that guarantees the competent performance evaluation. Transfer learning saves the system time on training and consumes fewer medical images to train than does training a model with no transfer learning. The lightweight architecture enables quicker inference, and is therefore appropriate to real-time clinical support systems. This would be a cost-efficient and scalable solution that would be able to help radiologists in timely detecting brain tumors. In general, the suggested system provides a balance between accuracy and efficiency to be used in practice in the context of healthcare systems with resource constraints.

B. System Architecture

It is a proposed system that includes MRI images of the brain in the preprocessing and normalization of homogenous input. Techniques of data augmentation to improve system learning through alleviating overfitting during training are implemented. MobileNetV2 will be used in extracting the features which are pertinent to the images and a fully connected layer will be used to classify the images into tumor or non-tumor images.

C. Key Advantages

The model size and complexity of the proposed system is smaller than the current approaches. It can be run effectively on systems based on CPU without using GPUs. It is also practical to use in medical application because it requires less time to predict and the system can be easily incorporated into current diagnostic systems.

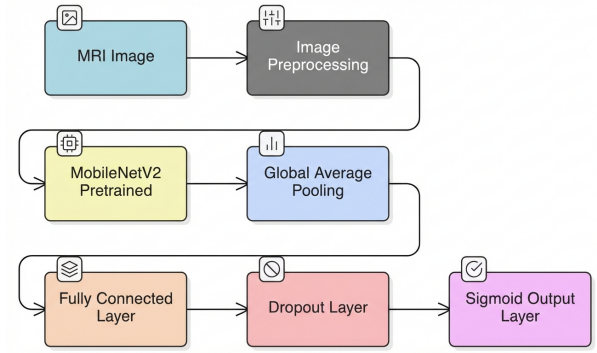


Fig. 1. System layout

V. METHODOLOGIES

Brain tumor detection system design will be a methodic process based on deep learning techniques of MRI brain images. Generally, the architecture of the design of the brain tumor detector system is illustrated using Fig. 2 and is a step-by-step process as follows:

A. MRI Brain Image Dataset

The system also starts with the collection of an MRI brain image dataset of tumorous and non-tumorous images. These images are the huge contribution to this system and the source of the images that are needed to determine the tumorous areas.

B. Dataset Extraction and Folder Structuring

First of all, it is sorted and divided into neatly labeled folders according to the names of the classes. It will ensure that all categories of its information respectively will be placed in their own folder hence will be readily loaded when training and validating the model will be carried out to the corresponding images and the respective labels.

C. Image Preprocessing

Image preprocessing is done to ensure that all the examples are uniform. All MRI images are resampled to 128x 128 pixel resolution in order to reduce computing resources and offer matching input dimensions. Stability is also provided by normalizing pixel intensities.

D. Data Augmentation

To increase diversity of the dataset and decrease overfitting various data augmentation methods are implemented. Therefore, it produces additional training samples that are rotated, inverted, zoomed, and translated and enables the model to be trained on strong and robust features.

E. Train-Validation Split

The processed and enhanced dataset is further divided into two parts: training and validation in 80:20 proportion. Representation of the training set will be to learn the model parameters, whereas the validation set will offer some form of monitoring performance and hence prevent overfitting.

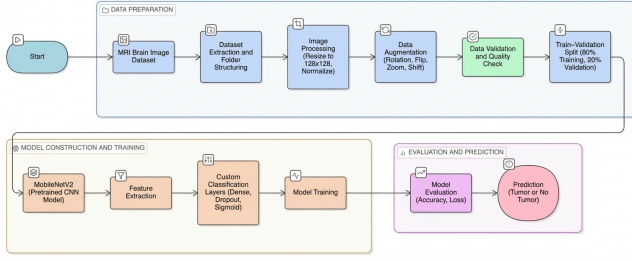


Fig. 2. Workflow of the proposed MobileNetV2-based brain tumor detection system using MRI images

F. Feature Extraction Using MobileNetV2

The pretrained model used in the work is the core feature extractor that is a MobileNetV2 model. This model has been chosen due to its efficiency and its capability of doing depth-wise separable convolutions, which have showed great results in the given task due to its ability to be constrained by the limitations of computation.

G. Custom Classification Layers

These extracted features are topped with custom classification layers. It is a stack of dense layers, or fully connected layers that have dropout and a sigmoid activation function that is used to either classify it as tumor or nontumor.

H. Model Training

The entire model is trained using the training set. During the process, the model learns how to recognize important patterns assisting in differentiating between tumor and normal images on the MRI.

I. Model Evaluation

One can then determine the efficiency of the new designed system by checking the model performance on the model accuracy and loss of the validation set after the model training procedure is done.

J. Prediction

The final step involves classifying the samples of the hidden images of MRI using the trained model. The output provided by the system either identifies the input image as a tumor or no tumor which is very instrumental in successful automated procedure of brain tumor detection.

VI. EXPERIMENTAL RESULTS

The current paper presents the experimental research of the proposed method of brain tumor detection using MobileNetV2. The suggested method is evaluated by means of quantitative values and qualitative outcomes of brain tumor prediction on unknown MRI scans.

A. Quantitative Results

The work of the suggested model was also evaluated according to the more traditional parameters such as accuracy and loss function. It was also noted that during training processes, the model is converging gradually because the values of its losses decrease over time and its accuracy also increases with each epoch. The last model was capable of classifying images successfully with a high level of accuracy when it was tested on the validation set.

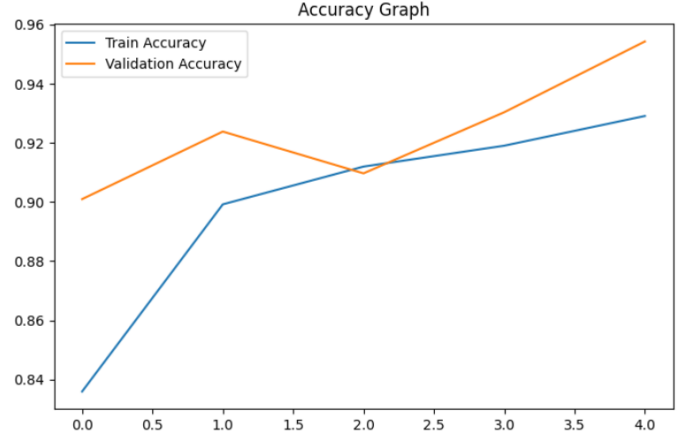


Fig. 3. Training and validation accuracy graph

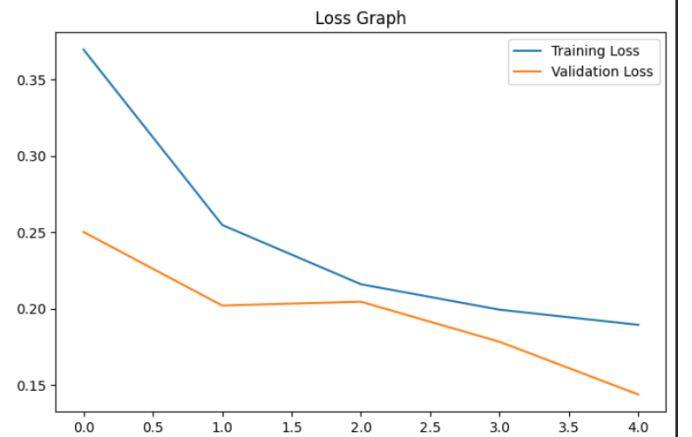


Fig. 4. Training and validation loss graph

B. Prediction Results

MRI images which were not previously seen were used to test the trained model to gauge its performance in the real world. It was able to categorize the scans with high scores of confidence of either tumor or non-tumor. These values indicate that this model has acquired meaningful and discriminative attributes of the training data. The fact that the system was very reliable is shown by the consistency in performance on new images. All in all, the process exhibits excellent generalization capabilities in real world brain tumor detection tasks.

TABLE I
PREDICTION RESULTS FOR DATASET 1

Image Name	Predicted Class	Confidence
img5.jpg	Brain Tumor	0.9995
img6.jpg	Brain Tumor	0.9923
img7.jpg	Healthy	0.9904
img8.jpg	Healthy	0.9990

TABLE II
PREDICTION RESULTS FOR DATASET 2

Image Name	Predicted Class	Confidence
img1.jpg	Brain Tumor	0.9979
img2.jpg	Brain Tumor	0.9998
img3.jpg	Healthy	0.9906
img4.jpg	Healthy	0.9981

TABLE III
PREDICTION RESULTS FOR DATASET 3

Image Name	Predicted Class	Confidence
img9.jpg	Brain Tumor	0.9998
img10.jpg	Brain Tumor	0.9592
img11.jpg	Healthy	0.9283
img12.jpg	Healthy	0.9948

VII. CONCLUSION

This paper proposed an automated brain tumor detection system, which operates on convolutional neural networks, to analyze the MRI image. The recommended solution was a movement of lightweight MobileNetV2 architecture to transfer learning in order to obtain correct classification without significant computational complexity. This renders the system appropriate to real-time and resource limited medical settings.

The results of the experiment proved the model to be effective in differentiating between tumor and non-tumor MRI scans at a high level of confidence and predictive accuracy on unknown test data. Transfer learning greatly saved time on training, although maintaining accuracy in classifications, which demonstrates that pretrained deep learning models can be effectively applied to medical image analysis problems.

Also, the proposed architecture is lightweight, which allows faster inference, which is essential to clinical decision-support systems. The model may help radiologists by offering a first-line automatic screening device, which will save diagnostic burden as well as limit the possibility of the human error.

Even though the proposed system performed well, it is currently limited to binary classification. The future work can build on this framework to produce multi-class tumor classification, tumor localization and segmentation and validation on bigger and more diverse MRI data to enhance generalization and clinical reliability.

Generally, the paper reiterates the possibility of the computer-aided diagnosis systems that are built using deep learning to improve the efficiency and accuracy of detecting brain tumors.

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